MIL-G-16265E
10 July 1973
SUPERSEDING
MIL-G-16265D
25 July 1963
(See 6.3)

MILITARY SPECIFICATION

# GASKETS, METALLIC-ASBESTOS, SPIRAL WOUND (FOR NAVY FLANGED JOINTS IN PIPING SYSTEMS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers gaskets of Navy design for use in Navy flanged joints in steam piping systems for pressures up to and including 600 pounds per square inch (psi) and temperatures up to and including 875°F.

1.2 Classification. Pipeline flange gaskets, with outer metal rings shall be of the following series, as specified (see 6.1):

Series 150.
Series 300.
Series 400.
Series 600.

## 2. APPLICABLE DOCUMENTS

# 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, SetUp.

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-17667 - Paper, Wrapping, Chemically Neutral (Non-Corrosive).
MIL-F-20670 - Flanges, Pipe, Carbon Steel, 150 PSI, W.S.P. (For Naval Shipboard Use).
MIL-A-24058 - Asbestos Packing and Gasketing Material, Impregnated.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

MILITARY

810-1385733 - Flanges, Steel For Steel Tubing Welding Neck.
810-1385809 - Fittings, Pipe, Cast Steel Flanged 300 PSI Maximum at 650°F Maximum.
5000-S4823-1385810 - Fittings, Pipe, Cast Steel Flanged.
5000-S4824-1385571 - Valves, Cast Steel Flanged Globe Angle and Cross.



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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code
Section VIII - Unfired Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A167-70 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Design (see 6.1).

3.1.1 Pipe line flange gaskets with outer metal rings. Gaskets shall consist of a refill snapped in a solid metal outer ring with a sliding fit. The ring shall have a thickness of 0.125 + 0.006 inch. The ring shall have a V-shaped groove or other satisfactory retaining feature machined on the inside periphery to fit the outside periphery of the refill. The design shall be such that the gasket can be compressed without damage to 0.119 inch thickness. Gaskets shall conform to tables I and II.

3.1.1.1 Refills for all series gaskets. Refills shall be of the inside and outside dimensions specified in tables I and II. The procuring activity is responsible for furnishing complete data including gasket material, dimensions, the compressed thickness, compression test load, the number and size of bolts, bolt material and service conditions. In those cases where this information cannot be obtained, the following design criteria may be utilized if desired for procurement of the gaskets. Gaskets shall be designed in accordance with the minimum design seating stress as defined in the ASME Boiler and Pressure Vessel Code, Section VIII, utilizing the formula $W_{m2} = 3.14\ bGy$ except that y be modified to read minimum 6000 psi.



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Table I - Gasket characteristics for series 150 Navy flanges - MIL-P-20670.[1/]

| Iron pipe size (ips) | Pipe outside diameter (od) | Gasket refills (see figure 1) | | Minimum number of plies | | | Outer[3/] metal ring od | Test load[4/] gasket pressure |
|---|---|---|---|---|---|---|---|---|
| | | | | Metal[2/] | | | | |
| | | | | Inside periphery ± 1/2 turn | Outside periphery ± 1/2 turn | Total (min.) | | |
| Inches | Inches | I.d.[5/] inches | O.d.[5/] inches | | | | Inches | Pounds |
| 1/4 | 0.540 | 3/4 | 1- 3/32 | 3 | 2 | 8 | 1- 5/8 | 11,340 |
| 3/8 | .675 | 7/8 | 1- 7/32 | 3 | 2 | 8 | 1- 3/4 | 11,340 |
| 1/2 | .840 | 1 | 1-11/32 | 3 | 2 | 8 | 1-15/16 | 11,340 |
| 3/4 | 1.050 | 1-1/4 | 1-21/32 | 3 | 2 | 9 | 2- 3/16 | 15,120 |
| 1 | 1.315 | 1-1/2 | 1-31/32 | 3 | 2 | 10 | 2- 5/8 | 15,120 |
| 1-1/4 | 1.660 | 1-3/4 | 2- 7/32 | 3 | 2 | 10 | 2- 7/8 | 15,120 |
| 1-1/2 | 1.900 | 2-1/4 | 2-23/32 | 3 | 2 | 10 | 3-7/16 | 22,680 |
| 2 | 2.375 | 2-3/4 | 3- 9/32 | 3 | 2 | 10 | 3-15/16 | 22,680 |
| 2-1/2 | 2.875 | 3-3/8 | 3-31/32 | 3 | 2 | 12 | 4- 1/2 | 22,680 |
| 3 | 3.500 | 3-7/8 | 4-15/32 | 3 | 2 | 12 | 5 | 30,240 |
| 3-1/2 | 4.000 | 4-3/8 | 4-31/32 | 3 | 2 | 12 | 5- 9/16 | 30,240 |
| 4 | 4.500 | 4-7/8 | 5-15/32 | 3 | 2 | 12 | 6- 1/16 | 30,240 |
| 4-1/2 | 5.000 | 5-3/8 | 6- 1/32 | 6 | 3 | 16 | 6- 9/16 | 37,800 |
| 5 | 5.563 | 5-7/8 | 6-19/32 | 6 | 3 | 16 | 7- 3/16 | 60,600 |
| 5-1/2 | 6.120 | 6-1/2 | 7- 7/32 | 6 | 3 | 16 | 7-11/16 | 60,600 |
| 6 | 6.625 | 7 | 7-25/32 | 6 | 3 | 17 | 8- 1/4 | 72,720 |
| 7 | 7.625 | 8 | 8-25/32 | 6 | 3 | 17 | 9- 3/8 | 72,720 |
| 8 | 8.625 | 9 | 9-27/32 | 6 | 3 | 18 | 10- 7/16 | 84,840 |
| 9 | 9.625 | 10 | 11- 3/32 | 6 | 3 | 22 | 11- 5/8 | 126,896 |
| 10 | 10.750 | 11 | 12- 3/32 | 6 | 3 | 22 | 12-11/16 | 135,960 |
| 11 | 11.750 | 11-1/2 | 12-19/32 | 6 | 3 | 22 | 14- 1/4 | 145,024 |
| 12 | 12.750 | 12-1/2 | 13-23/32 | 6 | 3 | 25 | 15- 5/16 | 163,152 |
| 14 od | 14.000 | --- | --- | - | - | -- | 16- 1/2 | 238,830 |
| 15 od | 15.000 | --- | --- | - | - | -- | 17- 1/2 | 251,400 |
| 16 od | 16.000 | --- | --- | - | - | -- | 18- 9/16 | 251,400 |
| 18 od | 18.000 | --- | --- | - | - | -- | 20- 5/8 | 276,540 |
| 20 od | 20.000 | --- | --- | - | - | -- | 22-11/16 | 396,120 |

1/ Gasket for 150 psi, 775°F. steam service when used with MIL-P-20670.

2/ There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.

3/ Id of outer metal ring equals od of refills plus 1/32 inch for all sizes.

4/ Corresponds to bolting load of 30,000 psi plus or minus 10 percent unit stress. Bolts; 3/4 inch diameter, 10 threads; 7/8 inch diameter, 9 threads; all other bolts 8 threads. When under this compressive test load, the gasket thickness shall be 0.130 ± 0.005 inch.

5/ Tolerance plus or minus 1/64 inch on the id and plus or minus 1/32 inch on the od.



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Table II - Series 300, 400, and 600 Navy flanges.

| Ips | Pipe od | Gasket refills (see figure 1) | | Minimum number of plies | | | Outer metal ring[1] od inch | | | Test load[6] gasket pressure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Metal[2] | | | [3]Series 300 | [4]Series 600 | [5]Series 600 | |
| Inches | Inches | Id inches[7] | Od inches[7] | Inside periphery ± 1/2 turn | Outside periphery ± 1/2 turn | Total (min) | | | | Pounds |
| 1/4 | 0.540 | 1/2 | 27/32 | 4 | 3 | 9 | 2- 1/8 | 1- 3/4 | 1- 3/4 | 12,600 |
| 3/8 | .675 | 3/4 | 1- 7/32 | 6 | 3 | 12 | 2- 3/8 | 2- 1/8 | 2- 1/8 | 12,600 |
| 1/2 | .840 | 3/4 | 1- 7/32 | 6 | 3 | 12 | 2- 3/8 | 2- 1/8 | 2- 1/8 | 12,600 |
| 3/4 | 1.050 | 1 | 1-17/32 | 6 | 3 | 14 | 2-11/16 | 2- 5/8 | 2- 5/8 | 20,200 |
| 1 | 1.315 | 1- 1/4 | 1-27/32 | 6 | 3 | 14 | 3- 1/8 | 2- 7/8 | 2- 7/8 | 20,200 |
| 1-1/4 | 1.660 | 1-11/16 | 2-11/32 | 6 | 3 | 14 | 3- 7/16 | 3- 1/4 | 3- 1/4 | 24,200 |
| 1-1/2 | 1.900 | 2 | 2-23/32 | 6 | 3 | 15 | 4 | 3- 3/4 | 3- 3/4 | 36,200 |
| 2 | 2.375 | 2- 9/16 | 3-11/32 | 6 | 3 | 15 | 4- 1/2 | 4- 5/16 | 4- 5/16 | 48,400 |
| 2-1/2 | 2.875 | 2-15/16 | 3-27/32 | 6 | 3 | 16 | 5- 3/16 | 5- 1/16 | 5- 1/16 | 72,500 |
| 3 | 3.500 | 3- 3/4 | 4-23/32 | 6 | 3 | 18 | 5- 3/4 | 5-13/16 | 5-13/16 | 72,500 |
| 3-1/2 | 4.000 | 4- 1/8 | 5- 7/32 | 6 | 3 | 20 | 6- 5/16 | 6- 5/16 | 6- 5/16 | 100,600 |
| 4 | 4.500 | 4- 3/4 | 5-27/32 | 6 | 3 | 20 | 6- 7/8 | 6-15/16 | 6-15/16 | 100,600 |
| 4-1/2 | 5.000 | 5- 5/16 | 6-15/32 | 6 | 3 | 22 | 7- 7/16 | 7- 9/16 | 7- 7/16 | 132,200 |
| 5 | 5.563 | 5-13/16 | 6-31/32 | 6 | 3 | 22 | 8 | 8- 5/16 | 8- 3/16 | 132,200 |
| 5-1/2 | 6.125 | 6- 3/8 | 7-19/32 | 6 | 3 | 22 | 8-11/16 | 8-13/16 | 8-15/16 | 150,900 |
| 6 | 6.625 | 6- 7/8 | 8- 7/32 | 6 | 3 | 24 | 9- 1/4 | 9-11/16 | 9- 9/16 | 198,400 |
| 6-1/2 | 7.125 | 7- 7/16 | 8-23/32 | 6 | 3 | 24 | 9- 7/8 | 10- 5/16 | 10- 3/16 | 198,400 |
| 7 | 7.625 | 7-11/16 | 9- 7/32 | 6 | 3 | 24 | 10- 7/16 | 10-13/16 | 10-11/16 | 262,000 |
| 7-1/2 | 8.125 | 8- 5/16 | 9-27/32 | 6 | 3 | 27 | 11- 1/8 | 11- 5/16 | 11- 5/16 | 262,000 |
| 8 | 8.625 | 8- 7/8 | 10-11/32 | 6 | 3 | 27 | 11-11/16 | 11-15/16 | 11-13/16 | 262,000 |
| 8-1/2 | 9.125 | 9- 5/16 | 10-27/32 | 6 | 3 | 27 | 12- 1/4 | 12- 5/16 | --- | 262,000 |
| 9 | 9.625 | 9- 7/8 | 11-15/32 | 6 | 3 | 32 | 12-13/16 | 12-13/16 | 13- 7/16 | 334,400 |
| 9-1/2 | 10.250 | 10- 5/16 | 11-31/32 | 6 | 3 | 32 | 13- 3/8 | 13- 9/16 | --- | 349,300 |
| 10 | 10.750 | 10-13/16 | 12-15/32 | 6 | 3 | 32 | 13-15/16 | 14- 1/16 | 15-11/16 | 445,900 |
| 10-1/2 | 11.250 | 11- 5/16 | 12-31/32 | 6 | 3 | 32 | 14- 1/2 | 14- 9/16 | --- | 349,300 |
| 11 | 11.750 | 11- 3/4 | 13-23/32 | 6 | 3 | 32 | 15- 3/16 | 15- 7/16 | 16-11/16 | 557,400 |
| 11-1/2 | 12.250 | 12- 5/16 | 14- 7/32 | 6 | 3 | 32 | 15- 3/4 | 15-15/16 | --- | 445,900 |
| 12 | 12.750 | 12- 7/8 | 14-23/32 | 6 | 3 | 32 | 16- 3/8 | 16- 7/16 | 17-15/16 | 557,400 |
| 14 od | 14.000 | 14- 1/4 | 15-31/32 | 6 | 3 | 32 | 17- 1/2 | 18-15/16 | --- | 557,400 |
| 15 od | 15.000 | 15- 1/4 | 16-31/32 | 6 | 3 | 32 | 18- 3/4 | 20- 3/16 | --- | 557,400 |
| 16 od | 16.000 | 16- 1/4 | 18- 7/32 | 6 | 3 | 35 | 19-13/16 | 21- 1/16 | --- | 557,400 |

1/ Id of outer metal ring equals od of refills plus 1/32 inch for all sizes.
2/ There shall be no asbestos plies between the metal plies on the inside and on the outside periphery.
3/ Gaskets for 300 psi steam service when used with Drawing 810-1385809 or Drawing 5000-S4824-1385571.
4/ Gaskets for steam service pressure 301 to 400 pounds per square inch gage (psig) and 850°F when used with Drawing 5000-S4823-1385810.
5/ Gaskets for steam service pressure 401 to 600 psig and 850°F, maximum temperature and boiler feed service pressure 551 to 750 psig and 425°F, maximum temperature, when used with Drawing 810-1385733 and valves for 600 pound service.
6/ Corresponds to bolting load to 25,000 psi unit stress for 1/4 to 1 inch inclusive and 30,000 psi for 1-1/4 inch and above.
7/ Tolerance plus or minus 1/64 inch on the id and plus or minus 1/32 inch on the od.

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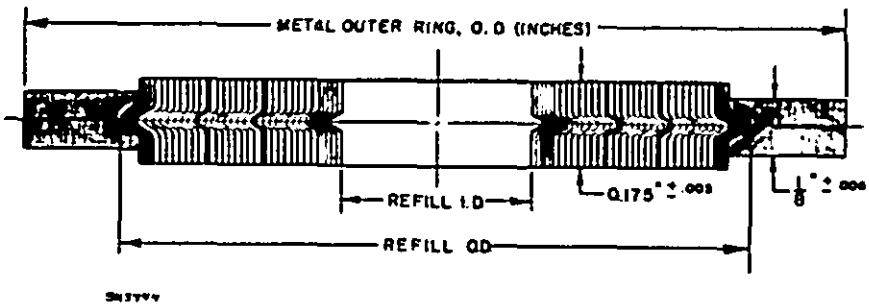



Figure 1 - Gasket.

3.2 Materials (see 6.1).

3.2.1 Metal strip. The metal strip used in the gasket shall be one of the following materials:

(a) Corrosion resisting steel AISI type 347, cold rolled and annealed, in accordance with ASTM A167-70.
(b) Corrosion resisting steel, AISI type 316L, in accordance with ASTM A167-70.
(c) Corrosion resisting steel, AISI type 309S, in accordance with the applicable requirements of ASTM A167-70 except that type 309S shall be stabilized with columbium (and tantalum not exceeding 0.10 percent) and their combination to be not less than 10 times the carbon content.

3.2.1.1 Special application. For special applications other materials may be used only upon specific approval of the activity concerned.

3.2.2 Metal outer ring. Unless otherwise specified in the contract or order, the metal outer ring shall be of soft carbon steel. At the option of the manufacturer a protective coating may be applied to the metal outer ring. The resulting dimensions shall conform to tables I and II.

3.2.3 Asbestos paper. Asbestos paper cushion shall be the low chloride variety conforming to type III of MIL-A-24058.

3.3 Construction. The gaskets shall be composed of a single strip of dovetail shaped metal spirally wound, starting on the inside and working outward ply by ply to the desired size. Between some of the plies shall be a cushion of asbestos paper (see tables I and II) either in a continuous single strip or in two strips depending upon the design.

3.3.1 Service. Complete gaskets and refills specified in tables I and II shall be satisfactory for temperatures up to 875°F.

3.3.2 Metal strip. The metal strip shall be not less than 0.007 and not more than 0.009 inch in thickness.

3.3.3 Gasket thickness. The thickness of the finished gasket (before compression) shall be 0.175 ± 0.005 inch.

3.3.4 Plies. A ply shall consist of one 360 degree turn of metal strip. Plies shall be counted adjacent to a terminal weld. An asbestos ply shall consist of one 360 degree turn of asbestos strip(s) depending upon the design.

3.3.5 Gaskets shall be capable of passing the compression load tests indicated in 4.4.1.



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3.3.6 Spot welds. The number of spacing of spot welds on the gasket id and od shall conform to the following:

(a) Terminal spot welds. The end of the terminal plies shall be spot welded to the adjacent plies with a minimum of 2 terminal spot welds. The first terminal spot weld shall be no greater than 1/4 inch from the end of the terminal ply. The second terminal spot weld shall be no greater than 3/4 inch from the first terminal spot weld measured along the perimeter. The terminal ply of the od is that ply which is spot welded to the adjacent ply to obtain the dimensions in tables I and II; a maximum of four additional unwelded plies of the same continuous metal strip may be wound over the spot welded ply to facilitate fitting the gasket to the outer metal ring.

(b) Additional spot welds on the terminal plies. Additional spot welds shall be located on the gasket id such that the total number of spot welds on the terminal ply (including the terminal spot welds described in (a) shall be not less than 3 and spaced at 3 inches maximum spacing along the perimeter between the adjacent spot welds in either direction.

3.4 Marking.

3.4.1 Each gasket shall be steel stamped clearly on one side of the outer metal ring to show the manufacturer's name, pipe size, the pressure series, material of construction and "N" to indicate that the gasket is for Navy flanges (example 2-600-SS-N).

3.4.2 Refills. When refills are purchased separate from the rings, each refill shall be identified by enclosing each refill in a paper or plastic bag, or by means approved by the procuring activity, having the manufacturer's name, pipe size, pressure series, material and "N" printed thereon. If a paper or plastic bag is used, it shall be adequately stapled to insure that the refill is retained in the bag. Identification markings need not be printed on plastic bags provided the marking is placed in the bag and is visible without requiring the bag to be opened.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All gaskets of the same series and size offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.2.2 Sampling for visual, dimensional and alloy examination. Sample gaskets shall be selected at random from each lot in accordance with table III for the examination specified in 4.3 and 4.4.4.

Table III - Sampling for visual, dimensional and alloy examination.

| Lot size (number of gaskets) | Sample size (number to be examined) | Acceptance number (Defectives) | Rejection number (Defectives) |
|---|---|---|---|
| 2 to 10 | All | 0 | 0 |
| 11 to 25 | 10 | 0 | 1 |
| 26 to 40 | 15 | 0 | 1 |
| 41 to 110 | 25 | 1 | 2 |
| 111 to 300 | 35 | 1 | 2 |
| 301 to 500 | 50 | 2 | 3 |
| 501 to 800 | 75 | 3 | 4 |
| 801 to 1300 | 110 | 4 | 5 |
| 1301 and over | 150 | 5 | 6 |



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4.2.3 Sampling for compression test. Sample gaskets shall be selected from each lot in accordance with table IV for the tests specified in 4.4.1. If any gasket fails the compression test, this shall be cause for rejection of the entire lot.

Table IV - Sampling for compression test.

| Lot size (Number of gaskets) | Sample size (Number to be tested) |
|---|---|
| 1 to 25 | 1 |
| 26 to 180 | 2 |
| 181 to 500 | 3 |
| 501 to 800 | 5 |
| 801 to 1,300 | 7 |
| 1,301 to 3,200 | 10 |
| 3,201 to 8,000 | 15 |
| 8,001 to 22,000 | 25 |
| 22,001 and over | 35 |

4.3 Visual and dimensional examination. Each of the sample gaskets selected in accordance with 4.2.2 shall be examined and measured to verify conformance with all of the requirements of this specification which do not include the compression tests. Examination shall be conducted as specified in table V. The number of plies shall be counted, a magnifying glass being used to insure accuracy of the count.

Table V - Classification of defects.

| Categories | Defects |
|---|---|
| Critical: | None defined. |
| Major: | |
| 101 | Series of gasket not as specified. |
| 102 | Number of metal plies less than specified minimum. |
| 103 | Spot welds (on inside and outside diameters) missing, not the required number, incorrectly spaced, or spot welds defective. |
| 104 | Thickness of metal strip not within the specified maximum and minimum values, or metal strip not continuous (one piece). |
| 105 | Uncompressed thickness of finished gasket not within specified tolerances. |
| 106 | Gasket refill not a sliding fit in the metal ring, or metal ring not provided with a satisfactory means of retaining the refill. |
| 107 | Inside or outside diameter of refill not within the specified tolerances. |
| 108 | Identification markings missing, illegible, incorrect or not as specified. |

4.3.1 Visual examination of spot welds. Spot welds shall be visually examined at a magnification of 5 to 10X. Defective spot welds, referenced in table V, category 103, shall be defined as welds showing burn-through or cracks exceeding 20 percent of the width of the metal strip.

4.4 Quality conformance inspection.

4.4.1 Compression load test.

4.4.1.1 The test shall consist of subjecting the gaskets to compression in a Baldwin Southwark, or equal, hydraulic compression machine of suitable capacity and measuring the thickness of the gaskets while under the tests loads specified in tables I and II, as applicable, for gaskets.

4.4.1.2 The gaskets shall be tested between steel test plates, the surfaces of which shall have a smooth finish machined with a circular lay (concentric or spiral) having a minimum of 32 serrations per inch of face width with depth of serrations not exceeding 0.003 inch of roughness height rating (RHR) finish of from 250 to 500. For special installations involving radioactive service or hazardous fluids where a finer finish is required, and so indicated in the contract or order, a maximum RHR of 125 shall be used. The test plates



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shall be centrally located under the ball and socket head of the test machine. The ball and socket head shall be used to equally distribute the load applied to the gasket. By means of inverted T straps, the upper compression plate shall be suspended and held in place when inserting a gasket. Four dial indicators graduated in 1/1000 inch divisions, located 90 degrees apart and mounted on the upper compression plate are employed for measuring the thickness of the gasket. (Electronic pick-up and recording of gasket thickness is acceptable in lieu of dial indicators.)

4.4.1.3 During the test, the increment of load shall be applied until a load corresponding to the test load in accordance with tables I and II is obtained for the gasket under test. The maximum rate of loading allowed will be 2,000 pounds per second. To insure reliability of gasket compression characteristics for the gaskets, the final compression test load shall be read just prior to the test plates making contact with the gasket compression stop ring (metal outer ring).

4.4.1.4 The steel test plates shall be placed in contact with each other and under an applied load of 3,000 pounds, the dial indicators shall be set so that a zero reading is obtained with the maximum travel of the indicator plunger. The plates shall then be separated, a gasket centrally located between plates and the upper plate brought into contact with the gasket but no load applied. The dial indicators shall then be read, and the average of the four readings should closely check the original micrometer thickness of the gasket. The load shall then be applied and the rate of loading depends on the size of gasket and the maximum load to be applied. The specified load shall be maintained only for a sufficient time, 5 to 10 seconds, to obtain the dial indicator readings. By means of reading the dial indicator scales counter clockwise instead of clockwise the actual thickness of the gasket shall be directly obtained. The average of the four dial indicator readings shall be considered to be the thickness of the gasket. After completion of the test, the load shall be released but the upper compression plate shall be left in contact with the gasket for approximately 10 seconds and the dial indicator readings shall be taken to determine the recovery of the gaskets. The test plates shall then be separated, the gasket removed from the test machine, measured for thickness by micrometers, examined for buckling on the inner periphery, condition of welds, and binding of the gasket filler in the outer metal ring. It may be noted that the final micrometer thickness and the released average dial indicator thickness of the gasket will be approximately the same providing the dial indicators are in proper adjustment. Occasionally between test runs the zero setting of the dial indicators shall be checked with the test plates in contact with one another. No buckling on the inner edge of the gasket shall be allowed (see 4.4.2(a)). Gaskets shall at least recover 0.010 inch over the compressed value obtained in the compression test specified in 4.4.1.3. It shall be the responsibility of the gasket manufacturer to adjust dial indicator readings to compensate for any deflection of test plates under load.

4.4.2 Test failures. Test failures are defined as follows:

(a) Gasket buckled excessively on inner periphery to the extent that metal plies are separated.
(b) Welding failed.
(c) Gasket thickness recovered less than 0.010 inch after release of compression test load.

4.4.3 Chemical analysis. Mill certification of materials may be accepted in lieu of performing chemical and physical tests. Such certification should contain the results of the chemical check analysis of the heat from the steel obtained, the heat number and date. When considered necessary, the samples selected in accordance with 4.2 shall be subjected to a chemical analysis to determine conformance with 3.2.1.

4.4.4 Alloy identity. Prior to packaging, each of the sample gaskets selected in accordance with 4.2.2 shall be examined for alloy identity by a spot tester manufactured by Systems Scientific Laboratory or Spot Tester, Inc., or other tester approved by the procuring activity. If any sample fails the alloy identity examination, the entire lot shall be rejected.

4.5 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

## 5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.2.)



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5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. Gaskets and refills, quantity as specified (see 6.1), shall be packaged in boxes conforming to PPP-B-566, PPP-B-636, PPP-B-665, or PPP-B-676, at the option of the supplier. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.1.1.1 Bundling. When specified (see 6.1), gaskets shall be bundled in the quantities and tied as follows:

5.1.1.1.1 Gaskets (with outer metal rings). Gaskets (with outer metal rings) shall be bundled as follows:

| Size (od) | Quantity and tying |
|---|---|
| Up to and including 5 inches. | 25 pieces. |
| Over 5 inches to 8 inches, inclusive. | 25 pieces, tied in 2 equidistant places. |
| Over 8 inches to 11 inches, inclusive. | 25 pieces, tied at 4 equidistant places. |
| Over 11 inches to 16 inches, inclusive. | 10 pieces, tied at 4 equidistant places. |
| Over 16 inches. | 5 pieces, tied at 4 equidistant places. |

Wrapping paper shall conform to MIL-P-17667.

5.1.1.2 Bulk. When specified (see 6.1), gaskets of like size and description shall be bulk packed directly into shipping containers for the level specified.

5.1.2 Level C. Packaging of gaskets and refills shall afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The supplier's normal packaging methods may be utilized when such meet the requirements of this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Gaskets and refills, packaged as specified (see 5.1), shall be packed in fiberboard boxes conforming to PPP-B-636, or PPP-B-640, class 2. Alternately, gasket and refills may be packed in containers conforming to any one of the following specifications at the option of the supplier.

| Specification | Type or class |
|---|---|
| PPP-B-585 | Class 3 use |
| PPP-B-591 | Class II, weather-resistant |
| PPP-B-601 | Overseas type |
| PPP-B-621 | Class 2 |
| PPP-B-576 | Class 2 |

Shipping containers shall have caselners conforming to MIL-L-10547. Caselners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caselners for weather-resistant or class 2 fiberboard boxes conforming to PPP-B-636 or PPP-B-640 respectively, may be omitted provided all center and edge seams and manufacturer's joints are waterproofed in accordance with the appendix to PPP-B-636. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto, except that fiberboard boxes shall be reinforced with pressure-sensitive, reinforced, filament tape applied in accordance with the appendix to the applicable fiberboard box specification, in lieu of steel strapping. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.



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# 5.2.2 Level B. Gaskets and refills, packaged as specified (see 5.1), shall be packed in fiberboard boxes conforming to PPP-B-636, grade V3c or PPP-B-640, class 1. Alternatively, gaskets and refills may be packed in containers conforming to any one of the following specifications at the option of the supplier:

| Specification | Type of class |
|---|---|
| PPP-B-585 | Class 1 or 2 use |
| PPP-B-591 | Class I, domestic |
| PPP-B-601 | Domestic type |
| PPP-B-621 | Class 1 |

Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds.

# 5.2.3 Level C. Gaskets and refills, packaged as specified (see 5.1), shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers packing or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules.

# 5.2.4 Filling of voids. Voids or cavities in the shipping containers, caused by wrapped bundles or bulk packing, shall be filled with cushioning materials to prevent shifting of contents and damage during handling, shipment and storage. The use of loose excelsior, shredded newspaper and similar materials which have not been refined for non-corrosiveness shall not be used unless these materials are enclosed in a sealed waterproof barrier material. Cushioning materials shall be non-corrosive to the packed item. Use of polystyrene "loose-fill" material is prohibited.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

## 6. NOTES

# 6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Series of gaskets required (see 1.2).
(c) Design requirements (see 3.1).
(d) Materials required (see 3.2).
(e) Quantities of gaskets and refills (see 5.1.1).
(f) If gaskets are to be bundled (see 5.1.1.1).
(g) If gaskets are to bulk packed (see 5.1.1.2).
(h) Levels of packaging and packing required (see 5.1 and 5.2).

# 6.2 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Custodians:
Army - MR
Navy - SH
Air Force - 82
Review activities:
Army - MR
Navy - SH, YD
DSA - IS
User activities:
Army - MU, ME
Navy - OS

Preparing activity:
Navy - SH
(Project 5330-0424)



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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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